INCH-POUND

MIL-PRF-44489
9 February 1996

PERFORMANCE SPECIFICATION

BEEF RAVIOLI IN MEAT SAUCE,

PACKAGED IN A FLEXIBLE POUCH, SHELF STABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance criteria for production of a finished product consisting of shelf stable beef ravioli in meat sauce, packaged in a flexible pouch, suitable for use by the Department of Defense as a component of operational rations. This component is consumed by combat personnel under worldwide environmental extremes as part of an operational ration, and is a source of nutritional intake. It is essential that this item be produced in accordance with good commercial practice to attain high standards of appearance, odor, flavor, and texture so that high levels of troop acceptance are achieved.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 4 and 5 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center, ATTN: SSCNC-WRE, Natick, MA 01760-5018 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-44073 - Packaging of Food in Flexible Pouches

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.1).

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC International

(Application for copies should be addressed to the AOAC International, 481, North Frederick Avenue, Suite 500, Gaithersburg, MD 20877.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.1), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Performance characteristics of the packaged food.

3.2.1 Commercial sterility. The packaged food shall be processed until commercially sterile.

3.2.2 Shelf life. The packaged food shall meet the minimum shelf life requirement of 36 months at 80°F or 6 months at 100°F.

3.2.3 Appearance.

3.2.3.1 Ravioli. The ravioli shall be uniform in size and shape.

3.2.3.2 Meat sauce. The meat sauce shall have a reddish brown color, shall contain identifiable pieces of meat, and may contain visible flecks of herbs and spices.

3.2.3.3 General. The finished product shall be a uniform mixture of beef ravioli in a Italian style flavored meat sauce. There shall be no bone or bone fragment in the packaged food. The packaged food shall be free of cartilage, coarse connective tissue, sections of tendons or ligaments and glandular material.

3.2.4 Odor and flavor.

3.2.4.1 General. The beef ravioli in meat sauce shall have a well balanced, typically Italian style odor and flavor.

3.2.4.2 Foreign. The packaged food shall be free from foreign odors and flavors such as, but not limited to, burnt, scorched, rancid, sour, or stale.

3.2.5 Texture.

3.2.5.1 Ravioli. The beef ravioli shall be moist and tender.

3.2.5.2 Meat sauce. The meat sauce shall be moderately thick.

3.2.6 Weight.

3.2.6.1 Net weight. The net weight of an individual pouch shall be not less than 8.0 ounces.

3.2.6.2 Drained weight.

3.2.6.2.1 Ravioli. The drained weight of a group of 12 to 16 intact beef ravioli in an individual pouch shall be not less than 4.0 ounces.

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3.2.6.2.2 Sauce meat. The drained weight of the sauce meat in an individual pouch shall be not less than 0.70 ounces.

3.2.7 Palatability. The finished product shall be equal to or better than the approved first article in palatability and overall appearance.

3.2.8 Nutrient content.

3.2.8.1 Water content. The water content of an individual pouch shall be not greater than 75.0 percent.

3.2.8.2 Protein content. The protein content of an individual pouch shall be not less than 6.2 percent.

3.2.8.3 Fat content. The fat content of an individual pouch shall be not greater than 4.0 percent.

3.2.8.4 Ash content. The ash content of an individual pouch shall be not greater than 2.0 percent.

3.2.8.5 Salt content. The salt content of an individual pouch shall be not greater than 1.5 percent.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article shall be inspected in accordance with the provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the performance characteristics or any appearance or palatability failure shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the methods of inspection of 4.5.1 through 4.5.8.

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4.4 Examination. The finished product shall be examined for compliance with the requirements specified in 3.2.3, 3.2.4, 3.2.5, 3.2.6, and 3.2.7. Defects and defect classifications are listed in table I.

TABLE I. Product defects 1/ 2/

Category		Defect
Major	Minor	
		Appearance
101		Bone or bone fragment measuring more than 0.3 inch in any dimension
102		Product not a uniform mixture as specified
	201	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material is more than 0.35 ounces
	202	Beef ravioli not uniform in size and shape
	203	Meat sauce appearance not as specified
		Odor and flavor
103		Odor and flavor not a well balanced Italian style
		Texture
	204	Beef ravioli not moist and not tender
	205	Meat sauce not moderately thick
		Weight
	206	Net weight not as specified
	207	Drained weight of beef ravioli not as specified
	208	Drained weight of sauce meat not as specified

1/ Presence of any foreign odors and flavors such as, but not limited to burnt, scorched, rancid, sour, or stale shall be cause for rejection of the lot.

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2/ Finished product not equal to or better than the approved first article in palatability and overall appearance shall be cause for rejection of the lot.

4.5 Methods of inspection.

4.5.1 Commercial sterility. Testing for commercial sterility shall be in accordance with MIL-PRF-44073.

4.5.2 Shelf life. Verification of shelf life shall be by contractor's certificate of conformance.

4.5.3 Appearance. Appearance is determined by compliance with 4.4.

4.5.4 Odor and flavor. Odor and flavor is determined by compliance with 4.4.

4.5.5 Texture. Texture is determined by compliance with 4.4.

4.5.6 Weight.

4.5.6.1 Net weight. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 ounce.

4.5.6.2 Drained weight. The pouch contents shall be poured into a flat-bottom container. A minimum of three times the pouch's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated so as to liquify rendered fat without breaking the beef ravioli and sauce meat pieces. The contents shall then be poured into a U.S. Standard No. 7 sieve in a manner that will distribute the product over the sieve without breaking the beef ravioli and sauce meat pieces. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4.5.7 Palatability. Palatability is determined by compliance with 4.4.

4.5.8 Nutrient content. The samples shall be prepared and analyzed for water content, protein content, fat content, ash content, and salt content in accordance with the methods of the Official Methods of Analysis of the AOAC. Test results shall be reported to the nearest 0.1 percent. Any result not conforming to water, protein, fat, ash, and salt requirements shall be cause for rejection of the lot.

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5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-PRF-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. When a first article is not required (see 3.1).

6.2 Ingredients. Ingredients may be as follows: Ravioli, tomato puree, beef, tomato paste, sugar, beef base, onions, salt, modified food starch, spices and flavorings.

6.3 Subject term (key word) listing.

Operational ration
Food processing
Pouched food

Custodians:

Army - GL
Navy - SA
Air Force - 35

Preparing activity:

Army - GL

(Project 8940-0843)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

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